

Computer Networks and Data Communication

Module 2

CS 212

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Jan-May 2026

Module wise CONTENTS:

- **Module 1 - Introduction:** An overview of networks, the Internet, services and protocols, RFCs and standards of Networking, Data communication & its components, types of data communication and communication channels. Key components of computer network; network devices; types of computer networks; network topology; reference models (Network Architectures) – OSI & TCP/IP.
- **Module 2 - Physical Layer:** Introduction and functions; types of transmission media – guided and unguided; integrated services digital network (ISDN); asynchronous transfer mode (ATM).
- **Module 3 - Data Link Layer:** Overview; sliding window protocols - Stop-&-Wait; Go-Back-N, Selective Repeat; Error detection and correction; Hamming Code, LRC, VRC, CRC, Checksum, Round Trip Time, Latency (Delay); Medium Access Sublayer: medium access control (MAC) Addresses; Switching techniques - circuit, message & packet; Random Access Protocols - Pure ALOHA, Slotted ALOHA, CSMA / CD, CSMA / CA; Controlled Access Protocols – FDMA, TDMA, CDMA; Channelization Protocols – Reservation, Polling, Token Passing; Ethernet (802.3 IEEE standard), Wireless Links, Wireless LAN - WiFi (802.11).
- **Module 4 - Network Layer:** Introduction; address resolution protocol (ARP); internet protocol (IP); internet control message protocol (ICMP); internet group management protocol (IGMP); IPv4 addressing, representation of IPv6; IPv4 vs IPv6; Conversion IPv4 to IPv6; Routing Protocols - routing information protocol (RIP), open shortest path first (OSPF); border gateway protocol (BGP); Enhanced Interior Gateway Routing Protocol (EIGRP).
- **Module 5 - Transport Layer:** Introduction; transport layer protocols - UDP & TCP, TCP connection establishment; TCP connection termination; TCP congestion control. **Application Layer:** Introduction – features & functions; application layer protocols; Web and HTTP, Email, P2P network, its types & applications.
- **Module 6 – Network Optimization, Mathematical Models, Introduction Advance Computer Networks, MANET, WSN, PCN, SDN, IoT.**

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- **Module 2 - Physical Layer:** Introduction and functions; types of transmission media – guided and unguided; integrated services digital network (ISDN); asynchronous transfer mode (ATM).

Physical Layer: Introduction

- **Physical Layer:** The physical Layer is the bottom-most layer in the Open System Interconnection (OSI) Model which is a physical and electrical representation of the system.
- It consists of various network components such as power plugs, connectors, receivers, cable types, etc.
- The physical layer sends data bits from one device(s) (like a computer) to another device(s).
- The physical Layer defines the types of encoding (that is how the 0's and 1's are encoded in a signal).
- The physical Layer is responsible for the communication of the unstructured raw data streams over a physical medium.

Physical Layer: Functions

- Functions performed by Physical Layer
- The physical layer maintains the data rate (how many bits a sender can send per second).
- It performs the Synchronization of bits.
- It helps in Transmission Medium decisions (direction of data transfer).
- It helps in Physical Topology (Mesh, Star, Bus, Ring) decisions (Topology through which we can connect the devices with each other).
- It helps in providing Physical Medium and Interface decisions.
- It provides two types of configuration Point to Point configuration and Multi-Point configuration.
- It provides an interface between devices (like PCs or computers) and transmission medium.
- It has a protocol data unit in bits.
- Hubs, Ethernet, etc. device is used in this layer.

Physical Layer: Functions

- This layer comes under the category of Hardware Layers (since the hardware layer is responsible for all the physical connection establishment and processing too).
- It provides an important aspect called Modulation, which is the process of converting the data into radio waves by adding the information to an electrical or optical nerve signal.
- It also provides a Switching mechanism wherein data packets can be forwarded from one port (sender port) to the leading destination port.

Types of Transmission media

- **1. Guided Media:** In this transmission medium, the physical link is created using wires or cables between two or more computers or devices, and then the data is transmitted using these cables in terms of signals. **Guided media transmission of the following types:**
 - **1. Twisted pair cable:** It is the most common form of wire used in communication. In a twisted-pair cable, two identical wires are wrapped together in a double helix. The twisting of the wire reduces the crosstalk. It is known as the leaking of a signal from one wire to another due to which signal can corrupt and can cause network errors. The twisting protects the wire from internal crosstalk as well as external forms of signal interference. **Types of Twisted Pair Cable:**
 - **Unshielded Twisted Pair (UTP):** It is used in computers and telephones widely. As the name suggests, there is no external shielding, so it does not protect from external interference. It is cheaper than STP.
 - **Shielded Twisted Pair (STP):** It offers greater protection from crosstalk due to shield. Due to shielding, it protects from external interference. It is heavier and costlier as compared to UTP.

Types of Transmission media

- **2. Coaxial Cable:** It consists of a solid wire core that is surrounded by one or more foil or wire shields. The inner core of the coaxial cable carries the signal and the outer shield provides the ground. It is widely used for television signals and also used by large corporations in building security systems. Data transmission of this cable is better but expensive as compared to twisted pair.
- **3. Optical fibers:** Optical fiber is an important technology. It transmits large amounts of data at very high speeds due to which it is widely used in internet cables. It carries data as a light that travels inside a thin glass fiber. **The fiber optic cable is made up of three pieces:**
 - **Core:** Core is the piece through which light travels. It is generally created using glass or plastic.
 - **Cladding:** It is the covering of the core and reflects the light back to the core.
 - **Sheath:** It is the protective covering that protects fiber cable from the environment.

Types of Transmission media

- **2. Unguided Media:** The unguided transmission media is a transmission mode in which the signals are propagated from one device to another device wirelessly. Signals can wave through the air, water, or vacuum. It is generally used to transmit signals in all directions. **Unguided Media is further divided into various parts:**
 - **1. Microwave:** Microwave offers communication without the use of cables. Microwave signals are just like radio and television signals. It is used in long-distance communication. Microwave transmission consists of a transmitter, receiver, and atmosphere. In microwave communication, there are parabolic antennas that are mounted on the towers to send a beam to another antenna. The higher the tower, the greater the range.
 - **2. Radio wave:** When communication is carried out by radio frequencies, then it is termed radio waves transmission. It offers mobility. It consists of the transmitter and the receiver. Both use antennas to radiate and capture the radio signal.
 - **3. Infrared:** It is short-distance communication and can pass through any object. It is generally used in TV remotes, wireless mouse, etc.

Integrated services digital network (ISDN)

- **ISDN** is employed as the network in data-link and physical layers but commonly ISDN is often limited to usage to Q.931 and related protocols. These protocols introduced in 1986 are a set of signaling protocols establishing and breaking circuit-switched connections, and for advanced calling features for the user.
- ISDN provides simultaneous voice, video, and text transmission between individual desktop videoconferencing systems and group videoconferencing systems.
- **Principle of ISDN:**
 - To support switched and non-switched applications
 - To support voice and non-voice applications
 - Reliance on 64-kbps connections
 - Intelligence in the network
 - Layered protocol architecture
 - Variety of configurations

Integrated services digital network (ISDN)

- **ISDN Interfaces:** The following are the interfaces of ISDN:

Basic Rate Interface (BRI)

- There are two data-bearing channels ('B' channels) and one signaling channel ('D' channel) in BRI to initiate connections.
- The B channels operate at a maximum of 64 Kbps while the D channel operates at a maximum of 16 Kbps.
- The two channels are independent of each other. For example, one channel is used as a TCP/IP connection to a location while the other channel is used to send a fax to a remote location. In iSeries ISDN supports a basic rate interface (BRI).
- The basic rate interface (BRI) specifies a digital pipe consisting of two B channels of 64 Kbps each and one D channel of 16 Kbps. This equals a speed of 144 Kbps.
 - In addition, the BRI service itself requires an operating overhead of 48 Kbps. Therefore, a digital pipe of 192 Kbps is required.

Integrated services digital network (ISDN)

Primary Rate Interface (PRI)

- Primary Rate Interface service consists of a D channel and either 23 or 30 B channels depending on the country you are in. PRI is not supported on the iSeries.
- A digital pipe with 23 B channels and one 64 Kbps D channel is present in the usual Primary Rate Interface (PRI). Twenty-three B channels of 64 Kbps each and one D channel of 64 Kbps equals 1.536 Mbps.
- The PRI service uses 8 Kbps of overhead also. Therefore, PRI requires a digital pipe of 1.544 Mbps.

Broadband-ISDN (B-ISDN)

- Narrowband ISDN has been designed to operate over the current communications infrastructure, which is heavily dependent on the copper cable however B-ISDN relies mainly on the evolution of fiber optics. According to CCITT B-ISDN is best described as ‘a service requiring transmission channels capable of supporting rates greater than the primary rate.

Integrated services digital network (ISDN)

■ ISDN Services:

Bearer Services

- Transfer of information (voice, data, and video) between users without the network manipulating the content of that information is provided by the bearer network.
- There is no need for the network to process the information and therefore does not change the content.
- Bearer services belong to the first three layers of the OSI model. They are well defined in the ISDN standard. They can be provided using circuit-switched, packet-switched, frame-switched, or cell-switched networks.

Teleservices

- In this, the network may change or process the contents of the data. These services correspond to layers 4-7 of the OSI model. Teleservices rely on the facilities of the bearer services and are designed to accommodate complex user needs. The user need not be aware of the details of the process.
- Teleservices include telephony, teletex, telefax, videotex, telex, and teleconferencing. Though the ISDN defines these services by name yet they have not yet become standards.

Integrated services digital network (ISDN)

- ISDN Services:

Supplementary Service

- Additional functionality to the bearer services and teleservices are provided by supplementary services.
- Reverse charging, call waiting, and message handling are examples of supplementary services which are all familiar with today's telephone company services.

Asynchronous transfer mode (ATM)

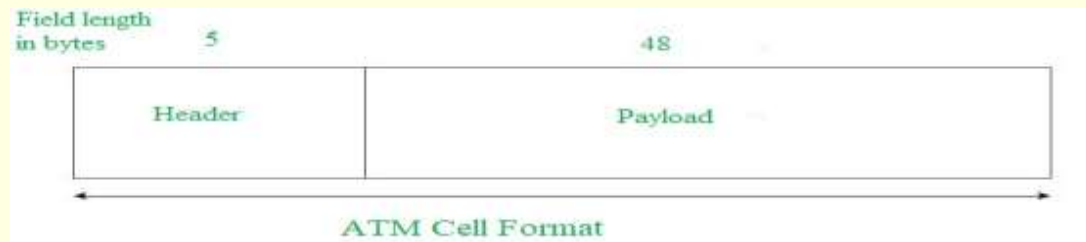
- Asynchronous Transfer Mode (ATM) is a networking technology that can be used in many real-world scenarios, including:
- **Telecommunications**
 - ATM is often used in the telecommunications industry because it can transmit voice, video, and data over the same network. It's also a good choice for commercial broadcasting due to its real-time transmission capabilities.
- **Wide Area Networks (WAN)**
 - ATM's high-speed data transmission capabilities make it a good choice for WANs, especially as the backbone network.
- **Multimedia streaming, video conferencing, and telecommunications infrastructure**
 - ATM's ability to integrate voice, video, and data on a single network, along with its prioritization features for real-time applications, makes it a preferred choice in these environments.
- **ATM is a connection-oriented network for cell relay that encodes data into small, fixed-size cells so that they can be transmitted over a physical medium.** It's a streamlined protocol with minimal error and flow control capabilities, which allows it to operate at high data rates.

Asynchronous transfer mode (ATM)

- **Asynchronous Transfer Mode (ATM):** It is an International Telecommunication Union- Telecommunications Standards Section (ITU-T) efficient for call relay and it transmits all information including multiple service types such as data, video, or voice which is conveyed in small fixed-size packets called cells.
- Cells are transmitted asynchronously and the network is connection-oriented.
- ATM is a technology that has some event in the development of broadband ISDN in the 1970s and 1980s, which can be considered an evolution of packet switching.
- Each cell is 53 bytes long – 5 bytes header and 48 bytes payload. Making an ATM call requires first sending a message to set up a connection.

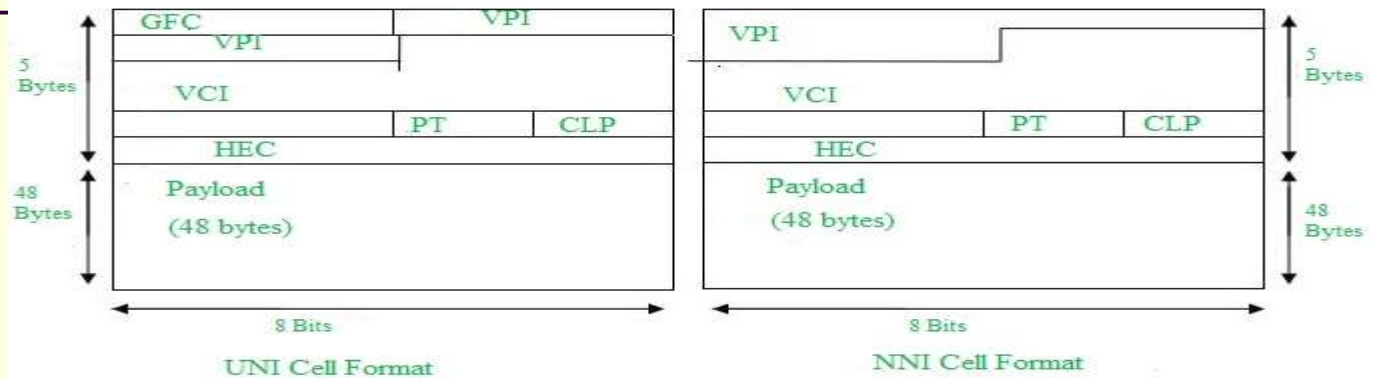
Asynchronous transfer mode (ATM)

- Subsequently, all cells follow the same path to the destination.
- It can handle both constant rate traffic and variable rate traffic. Thus, it can carry multiple types of traffic with end-to-end quality of service.
- ATM is independent of a transmission medium, they may be sent on a wire or fiber by themselves, or they may also be packaged inside the payload of other carrier systems.
- ATM networks use “Packet” or “cell” Switching with virtual circuits. Its design helps in the implementation of high-performance multimedia networking.
- **ATM Cell Format**
- As information is transmitted in ATM in the form of fixed-size units called cells. As known already each cell is 53 bytes long which consists of a 5 bytes header and 48 bytes payload.



Asynchronous transfer mode (ATM)

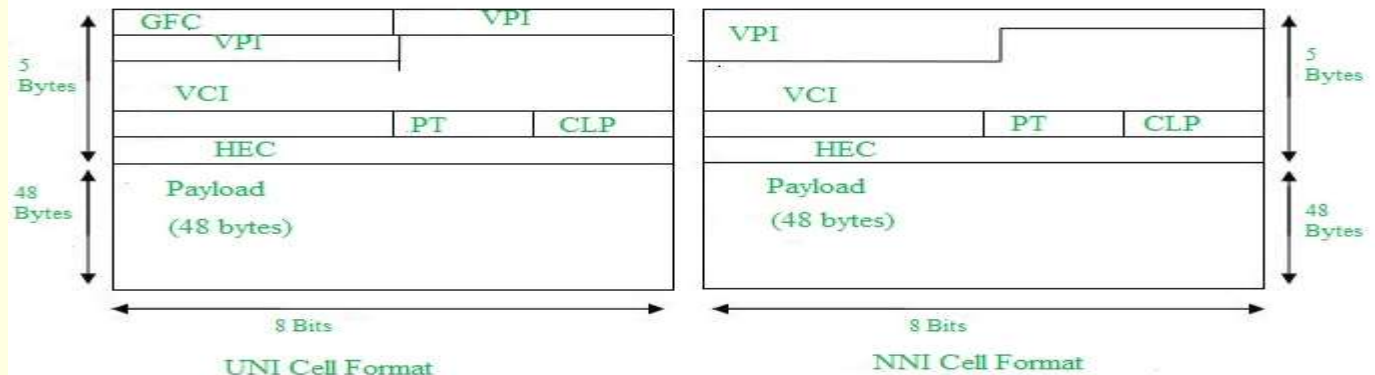
- Asynchronous Transfer Mode can be of two format types which are as follows:



- **UNI** – Universal Network Interface - for communication between ATM endpoints and ATM switches.
- **NNI** – Network to Network Interface - for communication between ATM switches.
- **GFC** – Generic Flow Control
- **VPI** – Virtual Path Identifier
- **VCI** – Virtual circuit Identifier – numerical value that identifies virtual circuit in computer network.
- **PT** – Penetrating Testing – to check for vulnerabilities.
- **CLP** – Cell Loss Priority – it indicates how a cell is treated if network is congested.
- **HEC** – High End Computing – anticipate future needs, overcome current challenges, and, effective use of system.

Asynchronous transfer mode (ATM)

- Asynchronous Transfer Mode can be of two format types which are as follows:



- **UNI (User Network Interface) Header:** This is used within private networks of ATMs for communication between ATM endpoints and ATM switches. It includes the Generic Flow Control (GFC) field.
- **NNI (Network to Network Interface) Header:** is used for communication between ATM switches, and it does not include the Generic Flow Control (GFC) instead it includes a Virtual Path Identifier (VPI) which occupies the first 12 bits.

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